

IPCC Tier 2 Livestock GHG Uncertainty Calculator

User Guide

CIAT / CGIAR Alliance — ClimateActionNetZero

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1 Overview

1.1 What this tool does

National GHG inventories based on IPCC Tier 2 methods require not only a central estimate of emissions but also a quantification of their uncertainty. This tool automates that process for livestock (cattle) using Monte Carlo simulation:

1. You supply your country-specific Tier 2 activity data and emission-factor inputs, each with an uncertainty range.
2. The tool randomly samples from the uncertainty distributions you have defined (thousands of times) and propagates those samples through the full IPCC Tier 2 equation chain (Eq. 10.1–10.34).
3. It reports the resulting 95% confidence interval around your emission estimate, identifies which parameters drive the most uncertainty (sensitivity analysis), and formats the results for submission in the format of IPCC 2006 Vol. 1 Ch. 3 Table 3.3.

The approach follows IPCC Approach 2 (Monte Carlo) as described in IPCC 2006 Vol. 1 Ch. 3.

1.2 Who this tool is for

This tool is designed for national inventory practitioners who:

- Are already conducting or have recently completed a Tier 2 cattle GHG inventory (enteric fermentation, manure management, direct and indirect N₂O from pasture deposition)
- Have at least rough estimates of the uncertainty in their key activity data and emission factors
- Need to report the uncertainty range for inclusion in national inventory reports

The tool assumes basic familiarity with IPCC Tier 2 methodology. Users do not need prior experience with Monte Carlo methods, but a conceptual understanding of probability distributions is helpful.

1.3 Emission sources covered

IPCC Category	Source
3.A.1	Enteric fermentation — cattle (CH ₄)
3.A.2	Manure management — cattle (CH ₄)
3.A.2	Manure management — cattle (N ₂ O direct)
3.A.2	Manure management — cattle (N ₂ O indirect)
3.C.4	Direct N ₂ O from pasture/range/paddock deposition
3.C.5	Indirect N ₂ O from pasture/range/paddock deposition

2 Preparing your data

2.1 What you need before you start

Before opening the tool, assemble the following:

1. **Activity data:** For each animal sub-category, the population (N), body weight (BW), weight gain (WG), milk yield (for dairy), milk fat %, digestible energy (DE%), manure management system (MMS) fractions, and other Tier 2 inputs.
2. **Uncertainty ranges:** For each parameter, an estimate of its uncertainty — typically either:
 - A percentage uncertainty (e.g. $\pm 20\%$ representing the half-width of a 95% confidence interval), or
 - Explicit lower and upper bounds.
3. **Manure management fractions** (if running manure N₂O): The percentage of manure allocated to each management system (pasture, solid storage, liquid, etc.).

If you do not yet have uncertainty estimates, see Section 2.4 for guidance on obtaining them.

2.2 Complete parameter reference

The table below lists all parameters accepted by the tool. Required parameters depend on the emission sources you are running. Parameters marked *n/a* in the IPCC default column must be supplied from your national data. The tool will auto-fill missing optional parameters with IPCC defaults and flag them in Tab 1 (see Section 3.2).

Table 1: Complete parameter reference — all Tier 2 inputs accepted by the tool.

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference	reference
Activity data							
N	Number of animals in this sub-category	head	<i>n/a</i>	±10%	Normal	Eq. 10.1	
BW	Average live body weight (also TAM = Typical Animal Mass)	kg	275	±15%	Normal	Eq. 10.3 / 10.6	
MW	Mature (adult) body weight — used in NEg growth equation	kg	300	±10%	Normal	Eq. 10.6	
WG	Average daily weight gain — set 0 for non-growing adults	kg/day	0	±30%	PERT	Eq. 10.6	
Milk	Daily milk yield per <i>lactating</i> animal (not sub-category average; tool multiplies by pct_calving internally). Set 0 for non-dairy sub-categories.	kg/head/day	4.0	±20%	Normal	Eq. 10.8	
Fat	Fat content of milk	%	4.0	±10%	Normal	Eq. 10.8	
pct_calving	Fraction of females that give birth in a year (0–1). Weights net energy for lactation, pregnancy, and N excretion.	fraction	0.60	±20%	Beta	2019 Ref. 10 p.20	Ch.
MilkPR	Protein content of milk — required for IPCC 2019 N excretion (Eq. 10.32A)	%	3.3	±10%	Normal	Table 10.11	
Energy coefficients							
DE	Digestible energy as % of gross energy — typical range 45–75%	%	55.0	±15%	Normal	Eq. 10.14	

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Cfi	Maintenance energy coefficient — depends on sex and lactation status	MJ/day/kg ^{0.75}	386	±30%	PERT	Table 10.4
Ca	Activity coefficient for locomotion energy	dimensionless	0.17	±30%	Triangular	Table 10.5
C	Growth coefficient for NEg equation — depends on sex (female 0.8, castrate 1.0, bull 1.2)	dimensionless	0.8	±30%	Triangular	Eq. 10.6
Cp	Pregnancy coefficient — 0.10 for pregnant animals	dimensionless	0.10	±10%	Beta	Table 10.7
hours	Daily working hours for draft animals — set 0 for non-draft	hours/day	0	±20%	PERT	Eq. 10.11
Tw	Mean daily temperature in winter (°C) — Cfi cold-climate adjustment. Leave blank or set 20 to disable.	°C	20	±25%	Normal	Eq. 10.2
Enteric fermentation						
Ym	Methane conversion factor (% of gross energy in feed converted to CH ₄)	%	6.5	±8%	PERT	Table 10.12
Manure management — CH₄						
Bo	Maximum CH ₄ producing capacity of manure	m ³ CH ₄ /kg VS	0.10	±20%	PERT	Table 10.16
ASH	Ash content of manure (fraction of dry matter)	fraction	0.08	±25%	PERT	Eq. 10.24

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
UE	Urinary energy as fraction of gross energy	fraction	0.04	±25%	PERT	Eq. 10.24
CP	Crude protein content of diet — used to estimate nitrogen excretion	%	10.0	±15%	Normal	Eq. 10.32
Manure management — N₂O (managed storage)						
EF3_S	N ₂ O emission factor for managed manure storage	kg N ₂ O- N/kg N	0.005	asymmetric	PERT	Table 10.21
Frac_GASMS	Fraction of managed manure N volatilised as NH ₃ /NO _x	fraction	0.20	asymmetric	PERT	2019 Table 10.22
EF4	N ₂ O EF for atmospheric N deposition	kg N ₂ O- N/kg N	0.010	asymmetric	Lognormal	Table 11.3
EF5	N ₂ O EF for N leaching/runoff	kg N ₂ O- N/kg N	0.0075	asymmetric	Lognormal	Table 11.3
Frac_LEACH_FF	Fraction of managed N lost through leaching	fraction	0.02	asymmetric	Lognormal	2019 Table 10.22
Manure management — N₂O (pasture/range/paddock)						
EF3_PRP	N ₂ O emission factor for dung/urine deposited on pasture	kg N ₂ O- N/kg N	0.02	asymmetric	PERT	Table 11.1
Frac_GASM_PRP	Fraction of N volatilised from dung/urine on pasture	fraction	0.21	asymmetric	PERT	2019 Table 11.3
Frac_LEACH_FF_PRP	Fraction of N lost through leaching from pasture deposition	fraction	0.30	asymmetric	PERT	2019 Table 11.3

Note on asymmetric uncertainty: Parameters flagged “asymmetric” use explicit lower/upper bounds from the IPCC tables rather than a symmetric $\pm\%$ formula. The template pre-fills these bounds automatically from the IPCC 2019 Refinement tables. *Note on pct_calving:* legacy column names `pct_lactating` and `pct_pregnant` are automatically renamed to `pct_calving` on upload.

2.3 Understanding the input structure

The tool uses an Excel template with a specific structure. Download the template from the **Data Input** tab (Tab 1) by clicking “Download Excel template”.

The template has the following key columns in the **Parameters** sheet:

Column	Description	Example
<code>cattle_type</code>	Broad cattle category	<code>dairy, non_dairy</code>
<code>aggregation_level</code>	Production system or region name	<code>Highland smallholder dairy</code>
<code>sub_category</code>	Animal sub-group	<code>cows, heifers, calves_female</code>
<code>parameter</code>	Parameter code (see Table 1)	<code>N, BW, Ym, DE, Bo</code>
<code>mean</code>	Central estimate (point value from your inventory)	<code>450 (BW in kg)</code>
<code>distribution</code>	Uncertainty distribution shape	<code>normal, pert, lognormal</code>
<code>uncertainty_pct</code>	Symmetric $\pm\%$ half-width of 95% CI	<code>20</code>
<code>lower</code>	Lower bound (alternative to <code>uncertainty_pct</code>)	<code>360</code>
<code>upper</code>	Upper bound	<code>540</code>
<code>param_type</code>	IPCC framing	<code>activity_data</code> or <code>coefficient</code>
<code>unit</code>	Unit of measurement	<code>kg, %, fraction</code>

Important: Each row is one parameter for one sub-category. A country with two production systems and five sub-categories each would have approximately 12–15 parameters $\times$ 10 sub-categories = 100–150 rows.

2.4 Defining sub-categories

Sub-categories should match your national inventory. Typical sub-categories include:

- Dairy: `cows` (lactating), `dry_cows`, `heifers`, `calves`, `bulls`
- Non-dairy beef: `breeding_cows`, `growing_males`, `heifers`, `calves`, `oxen`

Each sub-category needs its own rows for all relevant parameters. The values of `cattle_type`, `aggregation_level`, and `sub_category` together identify each group uniquely — they must be spelled consistently across all rows for the same group.

2.5 Obtaining uncertainty estimates

For activity data (N, BW, milk yields):

- **Survey-based estimates:** If populations are estimated from livestock surveys, the survey

sampling error provides a direct estimate of uncertainty. The 95% confidence interval from a survey translates directly to the **lower** and **upper** bounds (or to **uncertainty_pct** if symmetric).

- **Expert elicitation:** Ask the data provider to give a range within which they are 95% confident the true value falls.
- **IPCC defaults:** IPCC 2019 Refinement Chapter 11 provides uncertainty ranges for key emission factors. Use these as the starting point if no country-specific data is available.

For emission factors and coefficients (Ym, Bo, DE, MCF, etc.):

- **Literature review:** If you used a country-specific value from a published study, the study's reported confidence interval or standard deviation can be used.
- **IPCC default uncertainty ranges:** IPCC 2019 Refinement provides $\pm\%$ ranges for key EFs. These are reflected in the pre-filled columns in the template.
- **Expert elicitation:** Ask the expert who provided the value to estimate the plausible range.

If you genuinely have no information about a parameter's uncertainty, a reasonable default for emission factors is $\pm 20\text{--}40\%$. For population data with census-based estimates, $\pm 5\text{--}15\%$ is typical.

2.6 Choosing a distribution

Distribution	Shape	Best for
Normal	Symmetric bell curve	Activity data (N, BW) with symmetric measurement errors
PERT	Smooth bounded curve	Emission factors with a min, mode, and max
Triangular	Peaked triangle	Clear most-likely value with known hard limits
Uniform	Flat (equal probability)	Conservative fallback when only bounds are known
Lognormal	Right-skewed	Strictly positive values with right-skewed uncertainty (Bo, EFs)
Beta	Bounded [0, 1]	Fractions (pct_calving, Cp)

Recommended defaults:

- Use **Normal** for population (N) and body weight (BW) — typically measured with roughly symmetric errors
- Use **PERT** or **Lognormal** for emission factors (Ym, Bo, MCF, EF3–EF5, Frac parameters) — bounded below at zero with right-skewed uncertainty
- Use **Beta** for fractions bounded to [0, 1] (pct_calving, Cp)
- Use **Uniform** as a conservative fallback when you only know the plausible range

3 Loading your data (Tab 1 — Data Input)

3.1 Using the example datasets

Two built-in example datasets are included to let you explore the tool without uploading your own data:

- **Country X (hypothetical dairy):** A fictional East African highland dairy system — 12 parameters, dairy/cows, including a synthetic 5-year time series.
- **Country Y (hypothetical pastoral):** A fictional semi-arid pastoral beef system — 11 parameters, non-dairy/breeding cows, including a synthetic 5-year time series.

Select either from the “Country / Example Data” dropdown. The tool loads the data immediately and you can proceed directly to Tab 2.

3.2 Uploading your own data

Step 1 — Choose an IPCC Guidelines version. Before downloading the template, select the IPCC version your inventory follows:

- **IPCC 2006:** Uses the eight original manure management systems (MMS). Choose this if your national inventory is based on the 2006 Guidelines.
- **IPCC 2019 Refinement:** Adds four additional MMS types (anaerobic digester/biogas, aerobic treatment, solid storage covered/compacted, burned for fuel) and uses updated emission factors and N excretion equations from the 2019 Refinement. Choose this for inventories aligned with the 2019 Refinement.

The MMS dropdown in the downloaded template is automatically filtered to show only the manure systems valid for the version you selected.

Step 2 — Download and fill in the template.

1. Click “Download Excel template” to get the pre-formatted workbook
2. Fill in the **Parameters** sheet following the structure in Section 2.3
3. Fill in the **Manure_Management** sheet if you are running manure N₂O sources
4. (*Optional*) Fill in the **Parameter_TimeSeries** sheet if you want correlations computed automatically from historical data
5. Save the file as .xlsx and upload it using the file picker

Step 3 — Review upload notifications.

After a successful upload, a green notification confirms how many parameters were loaded. Any warnings appear as amber notifications.

Missing parameters and auto-fill: If your upload does not include all required parameters, the tool will:

- Still load and display all parameters that *were* supplied — you can review them immediately

in Tab 2 and Tab 3.

- Automatically fill in missing parameters using IPCC default values (see the "IPCC default" column in Table 1).
- Show a **yellow warning notification** listing exactly which parameters were auto-filled and with what values (e.g. "*3 parameters auto-filled from IPCC defaults: $Y_m = 6.5$, $B_o = 0.10$, $DE = 55.0$* ").
- Flag those parameters as **Missing** (purple/grey) in the QA/QC tab (Tab 2).

You should review whether the IPCC defaults are appropriate for your country context and, where possible, supply country-specific values before your final simulation run.

Common upload issues:

- `cattle_type` must be `dairy` or `non_dairy` (lowercase, no spaces)
- Distribution names must match exactly: `normal`, `pert`, `triangular`, `uniform`, `lognormal`
- If `uncertainty_pct` is left blank, both `lower` and `upper` must be filled
- `param_type` must be `activity_data` or `coefficient`

4 Reviewing data quality (Tab 2 — QA/QC)

4.1 What the QA/QC checks do

The QA/QC tab runs automated checks on your parameter values and flags any that are potentially problematic. Five statuses are possible:

Status	Colour	Meaning
Pass	Green	Value is within the expected range for this parameter
Warn	Amber	Value deviates noticeably from the typical range — check and document
Fail	Red	Value is outside the plausible range — likely a data entry error
Info	Blue	Value differs from an older benchmark source
Missing	Grey	A required parameter was not found in your upload

4.2 What to do for each status

Pass: No action needed. The value is consistent with IPCC default ranges.

Warn: Review the flagged parameter. A Warn means your value is unusual but not impossible. Possible actions:

- Confirm the value is correct for your country context
- If it is country-specific and well-justified, document it in your inventory report
- If it is an error, correct it in your Excel file and re-upload

Fail: These represent likely data entry errors (e.g. a live weight of 50,000 kg, a DE% of 150%). Correct the value before proceeding.

Info: These flags cite older benchmark sources (Penman 2000, Monni 2007) that may be less relevant than IPCC 2019 Refinement values. For parameters such as EF3, EF4, EF5, and Frac values, check the IPCC 2019 Refinement Chapter 11 (Table 11.1) for the current recommended range. If your value falls within the 2019 range, the Info flag can be disregarded for reporting purposes.

Missing: If a parameter is flagged as missing, the tool has auto-filled it with an IPCC default value (shown in the Tab 1 notification). Review whether the default is appropriate for your country context and, if possible, supply a country-specific value.

4.3 Documenting deviations

For any Warn or Info flags where you are deliberately using a country-specific value that deviates from the default, document the justification in your national inventory report. IPCC expert reviewers will look for this documentation.

5 Reviewing uncertainty distributions (Tab 3 — Uncertainty)

5.1 The parameter table

Tab 3 shows all parameters loaded from your data as an editable table. Each row shows:

- The parameter name, unit, and current mean value
- The distribution type currently assigned
- The uncertainty_pct or lower/upper bounds
- A visual preview of the distribution shape

You can edit values directly in the table. Clicking a cell opens an edit field.

5.2 Quick-set buttons

Two quick-set buttons are available for rapid exploration:

- **Set all activity data to Normal $\pm 15\%$:** Sets N, BW, and other activity data parameters to a Normal distribution with $\pm 15\%$ uncertainty. Click again to undo.
- **Set all coefficients to PERT:** Sets emission factors and other coefficients to PERT distribution. Click again to undo.

These are useful for a first exploratory run. Refine individual parameters before your final analysis.

6 Correlations (Tab 4 — Correlations)

6.1 Why correlations matter

If multiple parameters tend to move together (e.g. larger animals also tend to produce more milk), ignoring this relationship underestimates the uncertainty in the total emission estimate. For most Tier 2 analyses, ignoring correlations (the default “No correlations” setting) is a reasonable simplifying assumption. IPCC Approach 2 guidance recommends including known correlations where they exist.

6.2 Automatic correlations from time-series data

If you included a `Parameter_TimeSeries` sheet in your upload, the tool can compute historical correlations automatically:

1. On Tab 4, select “From template (auto)” under Activity Data Correlations
2. The tool fits a Gaussian copula to the historical parameter data and uses the resulting correlation matrix for sampling
3. The matrix is displayed for your review — inspect it to confirm correlations are plausible

6.3 Manual correlation matrix

If you prefer to specify correlations manually:

1. Select "Manual (upload CSV)" under Activity Data Correlations
2. Upload a square, symmetric CSV matrix of correlation coefficients (−1 to +1, diagonal = 1)
3. The tool verifies positive-definiteness and adjusts the matrix if needed

6.4 IPCC-guidance preset

A built-in preset based on IPCC guidance is available. It applies the following well-documented structural pairs:

Parameter A	Parameter B	ρ
BW	MW	0.85
BW	WG	0.40
Milk	Fat	−0.30
Milk	pct_calving	0.20
DE	CP	0.50
DE	Ym	−0.40
Cfi	Ca	0.60
N	BW	0.30

All other pairs are set to zero. A single uniform ρ can also be applied across all emission factors to represent shared systematic bias.

7 Running the simulation (Tab 5 — Simulate)

7.1 Settings

Before clicking Run, configure:

Number of iterations: The number of Monte Carlo samples.

- **1,000:** Quick test run to check the model runs correctly. Do not use for reporting.
- **10,000:** Minimum for final reporting purposes. Tail percentiles (2.5th/97.5th) are reliable.
- **25,000–30,000:** Recommended for final reporting, especially when correlations are enabled or you have many sub-categories.

GWP version: Choose between AR5 (CH₄ GWP = 28, N₂O = 265) and AR6 (CH₄ GWP = 27.9, N₂O = 273). Select the version required by your national inventory reporting guidelines.

Random seed: Fixing the seed ensures results are exactly reproducible. Change the seed to verify convergence: if the 95% CI changes substantially between runs, increase the number of iterations.

Analysis mode: Single-year (most common) or Trend (multi-year time series).

Emission sources: Tick all sources applicable to your inventory. Unticked sources contribute zero emissions to the total.

7.2 Running the simulation

Click **Run Monte Carlo Simulation**. A progress bar shows the simulation steps. For 10,000 iterations and a single production system, typical run time is 15–60 seconds.

When complete, results appear in the lower section of the same tab.

If something is missing or not working: All settings (iterations, GWP, seed, emission sources) remain active in the upper section of Tab 5 — simply adjust any setting and click **Run Monte Carlo Simulation** again. The previous results are replaced automatically. If you need to change your input data (parameters, distributions, or correlations), go back to the relevant tab (Tab 1–4), make the change, then return to Tab 5 and re-run.

7.3 Convergence diagnostics

Click “Run Diagnostic” (magnifying glass button) after a simulation to assess whether enough iterations were run for stable results:

- **MCSE%** (Monte Carlo Standard Error %): Pass < 0.5%, Warn 0.5–1%, Fail ≥ 1%
- **Mean drift %:** How much the mean from the first half of iterations differs from the second half. Pass < 2%
- **CI drift %:** How much the 95% CI bounds shift between the two halves. Pass < 5%

- **Convergence trace:** Running mean and 95% CI plotted against iteration count — should stabilise before the end of the run

Note: this tool uses **independent Monte Carlo**, not MCMC. Each iteration is drawn independently — there are no chains, no warmup, and no burn-in.

8 Understanding results (Tab 5 — Results)

8.1 Headline metrics

After a successful run, five value boxes show:

- **Total CO₂eq (t)**: The mean of the simulated total emission estimate, in tonnes CO₂-equivalent
- **Enteric CH₄ (t)**: Mean enteric fermentation contribution
- **Manure CH₄ (t)**: Mean manure management CH₄ contribution
- **Pasture N₂O (t)**: Mean combined direct and indirect N₂O from pasture deposition
- **Total CV (%)**: Coefficient of variation = $SD / \text{mean} \times 100$. Note: the IPCC reporting metric is the % uncertainty (MoE%) shown in the IPCC Report tab.

8.2 Uncertainty decomposition (AD vs EF)

If “Run uncertainty decomposition (AD/EF/Combined)” was checked, the results include a breakdown:

- **AD uncertainty**: Driven by activity data only (N, BW, Milk, WG) with all emission factors fixed at their means
- **EF uncertainty**: Driven by emission factors and coefficients only with activity data fixed at means
- **Combined uncertainty**: Both sources varying together

If AD uncertainty dominates, improving livestock population surveys would most effectively reduce total uncertainty. If EF uncertainty dominates, investing in country-specific EF measurement studies would be more impactful.

8.3 By-system and by-category breakdown

- **By-system breakdown**: Mean, 95% CI, MoE%, and CV% for each production system and emission source
- **By reporting category (IPCC Table 3.3 layout)**: Same information grouped by IPCC source category (3.A.1, 3.A.2, 3.C.4, 3.C.5)

9 Sensitivity analysis (Tab 6)

9.1 Reading the tornado chart

The tornado chart ranks parameters by their contribution to total uncertainty. The longest bars indicate the parameters that, if better constrained, would most reduce the 95% CI.

Each bar is labelled with:

- **Cattle type and sub-category:** e.g. dairy | cows
- **Parameter name:** e.g. Y_m (methane conversion factor), N (population), BW (body weight)

Bars are coloured by reducibility:

- **Green:** Uncertainty can be reduced with better local data (surveys or measurement studies)
- **Grey:** Uncertainty is largely irreducible with national-level data (e.g. fundamental biochemical constants)

9.2 SRC vs PRCC

Two sensitivity methods are computed:

- **SRC** (Standardised Regression Coefficient): Linear sensitivity. Reliable when relationships are approximately linear.
- **PRCC** (Partial Rank Correlation Coefficient): Rank-based, captures non-linear relationships. More robust but less interpretable.

Both are shown; they typically agree on the top-ranked parameters. If they disagree significantly, the emission pathway may be strongly non-linear.

10 Generating the IPCC report (Tab 7 — IPCC Report)

10.1 IPCC Table 3.3

The IPCC Report tab shows your results formatted as IPCC 2006 Vol. 1 Ch. 3 Table 3.3. The three uncertainty columns show:

- **AD uncertainty (%)**: % uncertainty (MoE%) from activity data alone
- **EF uncertainty (%)**: % uncertainty (MoE%) from emission factors alone
- **Combined uncertainty (%)**: Overall % uncertainty (MoE%)

Definition of % uncertainty: Following IPCC Table 3.3 conventions, all values are expressed as the half-width of the 95% confidence interval divided by the mean, multiplied by 100.

Example: a Combined uncertainty of 35% means the 95% CI is approximately $\pm 35\%$ of the mean emission estimate.

10.2 Downloads

Three download formats are available:

- **Excel report (.xlsx)**: Full workbook with run settings, IPCC summary table, uncertainty metrics, and sensitivity results (SRC and PRCC)
- **CSV report**: The IPCC Table 3.3 formatted table only, for easy import into inventory reporting software
- **Word report**: Narrative summary suitable for inclusion in the uncertainty section of your national inventory report

11 IPCC reporting conventions

11.1 Where uncertainty results go in the national inventory report

IPCC 2006 Vol. 1 Ch. 3 §3.3 describes the standard approach for Approach 2 (Monte Carlo):

1. Report the 95% confidence interval for each major source category and for the national total
2. Present the results in Table 3.3 format (included in the IPCC Report tab downloads)
3. Describe the Monte Carlo approach used (number of iterations, distributions chosen, correlation assumptions)
4. Document any parameters where country-specific values were used instead of IPCC defaults, including the justification

The methodology PDF (available on the Resources tab) provides a complete technical description of the equations, sampling approach, and convergence assessment that can be referenced in your national inventory report.

11.2 What IPCC expert reviewers will check

Based on experience with IPCC expert reviews, reviewers typically focus on:

- Whether the uncertainty metric reported is the correct one (% uncertainty = MoE%, not CV%)
- Whether the IPCC Table 3.3 format and column conventions are followed
- Whether Approach 2 (Monte Carlo) is described sufficiently to be reproducible
- Whether the number of iterations is sufficient for stable results
- The basis for any country-specific emission factor values that deviate substantially from IPCC defaults

12 Worked example — Country X (hypothetical dairy)

To illustrate the workflow, the following example uses the Country X built-in dataset.

Step 1 — Load data: Select “Country X (hypothetical dairy)” from the dropdown on Tab 1. The data loads automatically (12 parameters, dairy smallholder system).

Step 2 — QA/QC: Go to Tab 2. All parameters should pass or show only Info flags. The Info flags for EF3/EF4/EF5 cite older benchmark sources; for this hypothetical example, the values match IPCC 2019 defaults and the flags can be disregarded.

Step 3 — Uncertainty: Go to Tab 3. Note that activity data (N, BW, Milk, WG, Fat) use Normal distributions and coefficients (Ym, DE, Bo, etc.) use PERT or lognormal distributions.

Step 4 — Correlations: Go to Tab 4. Select “From template (auto)” — the example dataset includes a 5-year time series from which a correlation matrix is computed automatically.

Step 5 — Simulate: Go to Tab 5. Set iterations to 10,000, GWP to AR6, seed to 42. Click “Run Monte Carlo Simulation”.

Step 6 — Results: After about 30 seconds, results appear. Note the 95% CI and MoE%. Click “Run Diagnostic” to confirm convergence.

Step 7 — Sensitivity: Go to Tab 6. The tornado chart shows which parameters drive most uncertainty. In this example, Ym and BW for cows are typically the dominant contributors.

Step 8 — Report: Go to Tab 7. Download the Excel report to see the full IPCC Table 3.3 output.

For questions, contact Lolita Muller (Alliance Bioversity-CIAT) or Andreas Wilkes. \ This document is a draft for internal review.